

Discussion: Cross-Domain Analysis of Cognitive Integrity

Discussion: Cross-Domain Analysis of Cognitive Integrity I

Our cross-domain analysis of ten critical sectors reveals that Goal Hijacking is not merely a linguistic exploit but a structural corruption of the OODA Loop [?]. In every case—from drone swarms operating at millisecond time scales to diplomatic agents spanning months of deliberation—the attack vector was a transient signal that hijacked the agent's **Orientation** phase, rewriting its Functional Requirements in real-time. This section synthesizes the cross-domain findings, identifies universal attack patterns, evaluates CIF mechanism coverage, and acknowledges limitations.

4.1 Cross-Domain Attack Pattern Taxonomy I

Three universal attack patterns emerge across the ten domains. Each pattern corresponds to a distinct manipulation of the Axiomatic Design Matrix [?]:

Pattern 1: FR Polarity Inversion. The adversary flips the sign of a Functional Requirement, transforming a minimization objective into a maximization objective (or vice versa). The diagonal element A_{ii} effectively changes sign. This is the most common pattern, appearing in five domains.

Pattern 2: Constraint Relaxation. The adversary degrades a hard safety constraint to a soft preference, reducing the magnitude of the corresponding diagonal element A_{ii} toward zero. The FR nominally persists but loses its binding force.

4.1 Cross-Domain Attack Pattern Taxonomy II

Pattern 3: Context Boundary Violation. The adversary causes information from one operational context to bleed into another, introducing off-diagonal coupling where none existed. An element A_{ij} (where $i \neq j$) appears in the Design Matrix.

Domain	FR Polarity Inversion	Constraint Relaxation	Context Boundary Violation
1. Rare Earth Min- ing	✓		
2. Nation- State Al- liances			✓

4.1 Cross-Domain Attack Pattern Taxonomy III

3.



Cyber-
Security

4.



Drone
Wars

5.



Sup-
ply
Chain

6.



Biowar-
fare

7.



Food
Secu-
rity

4.1 Cross-Domain Attack Pattern Taxonomy IV

8.	✓		
Trade Wars			
9.	✓		
Infras- truc- ture			
10.			✓
Fake News			
Total	5	2	3

4.1 Cross-Domain Attack Pattern Taxonomy V

The dominance of FR Polarity Inversion (5/10 domains) suggests that the most effective Goal Hijacking attacks do not disable safety mechanisms but *co-opt* them—turning the agent's own optimization capabilities against its intended purpose. This is consistent with the Active Inference perspective on conflict [?], where adversaries exploit the agent's drive to minimize free energy by manipulating its generative model.

4.2 The Independence Axiom Under Adversarial Pressure I

The Independence Axiom (sec:methodology) requires that Functional Requirements remain independent—i.e., the Design Matrix $[A]$ stays diagonal. Goal Hijacking violates this axiom by introducing off-diagonal terms, **coupling** the Instruction channel with the Data channel. When a drone reads “Hospital” (Data) as “Target” (Instruction), the design becomes Coupled. When a cyber-security agent’s “Prevent Access” FR is overridden by a fabricated “Restore Availability” urgency, independent FRs become entangled.

4.2 The Independence Axiom Under Adversarial Pressure II

The CIF defense strategy maps directly to restoring independence. Paper 1's defense composition algebra [?] provides the formal basis, with the recommended stack achieving 94–100% detection at the parametric design ceiling across 950 attack scenarios and four production architectures [?], and Paper 3 [?] operationalizes this stack through deployment guides, monitoring, incident-response playbooks, and cost–benefit frameworks. The key insight from our cross-domain analysis is that different domains require different defense compositions, but the *vocabulary* of defense mechanisms is universal—the five canonical CIF mechanisms established in sec:methodology suffice to address all ten domains.

4.3 OODA Transient Dynamics I

Traditional engineering assumes Functional Requirements are static. Cyber-cognitive warfare proves they are dynamic variables. The adversary's goal is to introduce a **fast transient**—a high-frequency change in the agent's goal state that executes faster than either the human supervisor's OODA loop or the system's defense mechanisms can detect.

This creates a fundamental **race condition** [?]: if the accumulation of evidence for the hijack takes longer than the action execution, the agent fails. The temporal dynamics vary enormously across domains:

Domain	OODA Cycle Time	Transient Duration	Defense V
Drone Wars	Milliseconds	Sub-second	Near-zero
Cyber-Security	Seconds	Seconds	Seconds
Infrastructure	Minutes	Minutes–Hours	Minutes
Supply Chain	Hours	Hours–Days	Hours

4.3 OODA Transient Dynamics II

Food Security	Days	Days–Weeks	Days
Trade Wars	Weeks	Weeks–Months	Weeks
Rare Earth Mining	Months	Months	Weeks–Months
Nation-State	Months–Years	Days–Months	Weeks
Biowarfare	Variable	Hours–Months	Hours (systemic)
Fake News	Minutes–Hours	Seconds–Days	Minutes

4.3 OODA Transient Dynamics III

CIF addresses the race condition through **Drift Detection** (S_{drift} , defined in sec:methodology): sudden orientation shifts are flagged regardless of whether the content passes semantic analysis. For fast-cycle domains (Drones, Cyber), this is supplemented by **Behavioral Invariants** that impose hard temporal dampers—mandatory latency on override commands that exceeds the characteristic duration of synthetic transients.

4.3 OODA Transient Dynamics IV

Recent empirical benchmarks substantiate the race condition analysis. The Agent Security Bench (ASB) evaluation [?], presented at ICLR 2025, measured an average attack success rate of 84.3% across 10 agent scenarios encompassing 400+ integrated tools and 27 distinct attack/defense methods. Critically, ReAct-prompted agents—the dominant architecture for tool-using LLMs—exhibited the highest vulnerability, suggesting that the chain-of-thought reasoning patterns that make agents capable also make them exploitable. The InjecAgent benchmark [?] corroborates this finding: GPT-4-based agents were vulnerable to indirect prompt injection 24% of the time in base conditions, with the vulnerability nearly doubling under reinforcement. These benchmarks validate the central claim of the transient dynamics analysis: defense mechanisms consistently fail to keep pace with attack execution speed when operating within the same cognitive loop.

4.3 OODA Transient Dynamics V

The temporal asymmetry is further illuminated by the distinction between *static* and *dynamic* prompt injection [?]. Static injections (pre-positioned payloads in data sources) create persistent coupling in the Design Matrix, while dynamic injections (real-time adversarial responses) introduce transient coupling that must be detected within a single OODA cycle. The ASB results show that current defense methods achieve only 19.7% average defense success rate—a ratio that underscores the inadequacy of content-based filtering alone and motivates CIF's architectural approach to defense composition.

4.3 OODA Transient Dynamics VI

The formalization of OODA transients as Design Matrix perturbations also reveals a connection to control theory: the CIF mechanisms function as a **low-pass filter** on the agent's goal state, attenuating high-frequency (adversarial) signals while preserving low-frequency (legitimate) updates. This damping function is what the original draft termed “Cognitive Damping”—more precisely described as the joint operation of Drift Detection and Behavioral Invariants.

4.4 CIF Mechanism Coverage Analysis I

A critical validation of the CIF framework is whether the five canonical mechanisms provide adequate coverage across diverse operational domains. The following matrix maps primary CIF defenses to the ten domains analyzed:

CIF Mecha- nism	RE	NS	Cy	Dr	SC	Bio	FS	TW	Inf	FN	Total
Cognitive Firewall ($\mathcal{F}$)				✓		✓				✓	3
Belief Sandbox- ing ($\mathcal{B}_{\text{prov}}$)		✓				✓	✓		✓		4
Behavioral Invari- ants	✓				✓	✓		✓	✓		5

4.4 CIF Mechanism Coverage Analysis II

Drift De- tection (S_{drift})	✓				✓	✓	3
Byzantine Consen- sus ($\mathcal{B}_{\text{con}}$)	✓		✓	✓			3

4.4 CIF Mechanism Coverage Analysis III

Key: RE=Rare Earth, NS=Nation-State, Cy=Cyber, Dr=Drone, SC=Supply Chain, Bio=Biowarfare, FS=Food Security, TW=Trade Wars, Inf=Infrastructure, FN=Fake News.

Key findings:

1. **Behavioral Invariants are the most universal mechanism** (5/10 domains), reflecting their role as the “last line of defense”—hard predicates that trigger regardless of semantic content.
2. **All five mechanisms appear in at least 3 domains**, confirming that the CIF vocabulary is neither redundant nor incomplete for the application space surveyed.
3. **No single mechanism suffices alone.** Every domain requires at least two mechanisms in composition, consistent with Paper 1’s defense-in-depth architecture [?].

4.4 CIF Mechanism Coverage Analysis IV

4. **Mechanism selection correlates with attack pattern.** FR Polarity Inversion domains predominantly use Behavioral Invariants (the inverted FR violates a hard predicate). Context Boundary Violation domains predominantly use Cognitive Firewall or Belief Sandboxing (the boundary enforcement prevents cross-context contamination).

4.5 Novel Defense Patterns I

While the five canonical CIF mechanisms provide comprehensive coverage, three domains introduce genuinely novel instantiations that extend the CIF vocabulary:

Verification Channel Separation (Biowarfare). The biowarfare domain's defense architecturally separates the *semantic* channel (text justification) from the *physical* channel (protein folding simulation). The verification module is literally “deaf” to the persuasive rhetoric of the prompt, making Goal Hijacking structurally impossible within the verification pathway [?, ?]. This pattern generalizes: any domain where physical simulation can independently verify claims should route verification through a semantics-free channel.

4.5 Novel Defense Patterns II

Active Perturbation Probing (Trade Wars). Standard Drift Detection passively monitors belief changes. The trade wars domain extends this to *active probing*: the agent deliberately injects small perturbations into its decision model to test whether observed correlations are robust or adversarial artifacts [?]. If a policy recommendation relies on a counter-intuitive correlation that vanishes under slight noise, it is flagged as a potential adversarial artifact. This is analogous to adversarial robustness testing in machine learning [?, ?], but applied at the decision-policy level rather than the input level.

4.5 Novel Defense Patterns III

Physics-Informed Invariants (Infrastructure). Standard Behavioral Invariants are domain-agnostic predicates. The infrastructure domain specializes these as *physics-informed invariants* that encode conservation laws (e.g., Kirchhoff's Laws: $\sum I_{\text{in}} = \sum I_{\text{out}}$) as runtime predicates [?]. This leverages the mathematical structure of the physical domain to create invariants that are provably unforgeable—an adversary cannot fabricate sensor data that simultaneously satisfies conservation laws and achieves the desired hijack, without also providing the energy budget that real physics would require.

4.6 Byzantine Fault Tolerance Validation I

Paper 1's Byzantine Consensus mechanism ($\mathcal{B}_{\text{consensus}}$) [?] drew on the classical BFT result that $n \geq 3f + 1$ honest nodes can tolerate f Byzantine (arbitrarily faulty) nodes [?]. At the time of Paper 1's publication, the application of BFT principles to AI agent safety was largely theoretical. Two independent 2025 research efforts have since provided empirical and formal validation.

4.6 Byzantine Fault Tolerance Validation II

Formal BFT-AI Isomorphism. deVadoss and Artzt [?] establish a formal connection between unreliable AI artifacts and Byzantine nodes, demonstrating that the mathematical framework of BFT directly applies to AI safety scenarios where individual agents may produce arbitrary (including adversarially manipulated) outputs. Their key contribution is the *isomorphism argument*: a multiagent system where f agents have been goal-hijacked is formally equivalent to a distributed system with f Byzantine nodes, and the classical fault tolerance guarantees transfer directly. This validates Paper 1's adoption of the $n \geq 3f + 1$ quorum requirement for CIF's Byzantine Consensus mechanism.

4.6 Byzantine Fault Tolerance Validation III

Emergent Byzantine Resistance in LLMs. Zheng et al. [?] investigate the reliability of LLM-based multiagent systems from a BFT perspective and report a surprising finding: LLM agents demonstrate “stronger skepticism” when processing messages that contain erroneous or contradictory information, compared to traditional software agents that process all inputs with equal trust. This emergent property—which the authors attribute to the instruction-following training that teaches models to identify inconsistencies—suggests that LLM-based agents may possess natural Byzantine-resistant properties that can be leveraged by CIF’s consensus mechanism.

4.6 Byzantine Fault Tolerance Validation IV

The implications for CIF are twofold. First, the deVadoss-Artzt isomorphism confirms that Paper 1's quorum formula is not merely an analogy but a formally justified bound: a multiagent system with n agents can tolerate f goal-hijacked agents if and only if $n \geq 3f + 1$, with the bound being tight. Second, the Zheng et al. finding suggests that CIF's Byzantine Consensus may be more effective in LLM-based systems than classical BFT would predict, because the “honest” agents are not merely following protocol but are actively skeptical of anomalous inputs. This represents a potential advantage of cognitive agents over traditional distributed systems, where honest nodes are presumed to be passive rule-followers.

The emergence of BFT for AI Safety as an active research area—evidenced by a dedicated 2025 workshop and multiple concurrent publications—independently validates the trajectory established by Paper 1's adoption of Byzantine consensus as a canonical CIF mechanism.

4.7 Comparison with Existing Frameworks I

The CIF-AD-OODA integration model exists within a rapidly evolving landscape of AI security frameworks. We compare with six established and emerging alternatives to clarify CIF's distinctive contributions and complementary relationships.

4.7 Comparison with Existing Frameworks II

OWASP Top 10 for Agentic Applications [?]. Released in December 2025, this standard designates **ASI-01: Agent Goal Hijack** as the #1 risk for deployed agentic AI systems—a direct validation of this paper’s central thesis. The OWASP taxonomy identifies ten vulnerability classes spanning prompt injection, insecure tool use, supply chain risks, and insufficient output validation. CIF complements OWASP by providing *formal defense mechanisms* with composable guarantees, whereas OWASP primarily catalogs threats and recommends mitigations without formal composition algebra. Notably, ASI-01 through ASI-10 map naturally onto CIF’s adversary taxonomy: ASI-01 (Goal Hijack) corresponds to the teleological corruption modeled throughout this paper, ASI-03 (Insecure Tool Integration) maps to Ω_2 peripheral vectors, and ASI-07 (Multi-Agent Manipulation) aligns with Ω_4 coordination attacks.

4.7 Comparison with Existing Frameworks III

MAESTRO Framework [?]. The Cloud Security Alliance's Multi-Agent Environment Security, Threat, Risk, and Outcome (MAESTRO) framework provides a layered threat modeling approach specifically designed for multi-agent architectures. MAESTRO identifies seven architectural layers (Foundation Model, Data Operations, Agent Core, Tool Integration, Multi-Agent Orchestration, Deployment, and Ecosystem) and maps threats to each layer. CIF's contribution relative to MAESTRO is the formal defense composition algebra: while MAESTRO enumerates threats per layer, CIF provides mechanisms that compose in series and parallel with provable detection guarantees. The two frameworks are complementary—MAESTRO identifies *where* threats emerge in the architecture, while CIF specifies *how* to defend against them formally.

4.7 Comparison with Existing Frameworks IV

MITRE ATLAS [?]. ATLAS provides an adversarial threat landscape specifically for AI systems, organized as a knowledge base of techniques and tactics analogous to ATT&CK for traditional cyber threats. CIF's adversary taxonomy (Ω_1 – Ω_5) is compatible with ATLAS's technique classification but adds the *structural* dimension of Design Matrix analysis and the *temporal* dimension of OODA transient dynamics. ATLAS describes *what* adversaries do; CIF additionally models *why* certain attacks succeed (Independence Axiom violation) and *how* to compose defenses (defense algebra).

NIST AI 600-1 [?]. The NIST Generative AI Profile identifies 12 risk categories specific to generative AI, including confabulation, information integrity, and CBRN information risks. CIF addresses the goal manipulation subset formally—what NIST categorizes as “information integrity” and “human-AI configuration” risks. The NIST framework provides risk governance guidance but does not specify runtime defense mechanisms; CIF fills this operational gap.

4.7 Comparison with Existing Frameworks V

ATFAA/SHIELD Framework [?]. Narajala and Narayan (2025) propose a nine-threat model for agentic AI systems with a corresponding defense architecture. Their threat model overlaps substantially with CIF's adversary taxonomy but uses a different organizational principle (threat type rather than access level). CIF's advantage is the formal connection to Axiomatic Design theory, which enables structural analysis of attack success conditions (Independence Axiom violation) rather than purely empirical threat enumeration.

4.7 Comparison with Existing Frameworks VI

Industry Safety Frameworks (Anthropic RSP [?], OpenAI Preparedness [?], DeepMind FSF [?]). These company-specific frameworks address training-time alignment through evaluation thresholds, red-teaming protocols, and capability elicitation testing. CIF operates at a complementary layer: *deployment-time cognitive integrity*. The industry frameworks ensure that a model is safe when deployed; CIF ensures that a deployed model remains safe under adversarial pressure in a multiagent environment. The distinction mirrors the difference between manufacturing quality control (training-time) and field maintenance (deployment-time) in traditional engineering.

4.7 Comparison with Existing Frameworks VII

The comparison reveals CIF's distinctive position: it is the only framework that integrates formal structural analysis (via AD), temporal dynamics (via OODA), and composable defense mechanisms into a unified model. Other frameworks provide either threat taxonomies without formal defenses (OWASP, ATLAS, NIST), layered architecture mapping without composition algebra (MAESTRO), or training-time alignment without deployment-time protection (industry frameworks). CIF's contribution is precisely this integration.